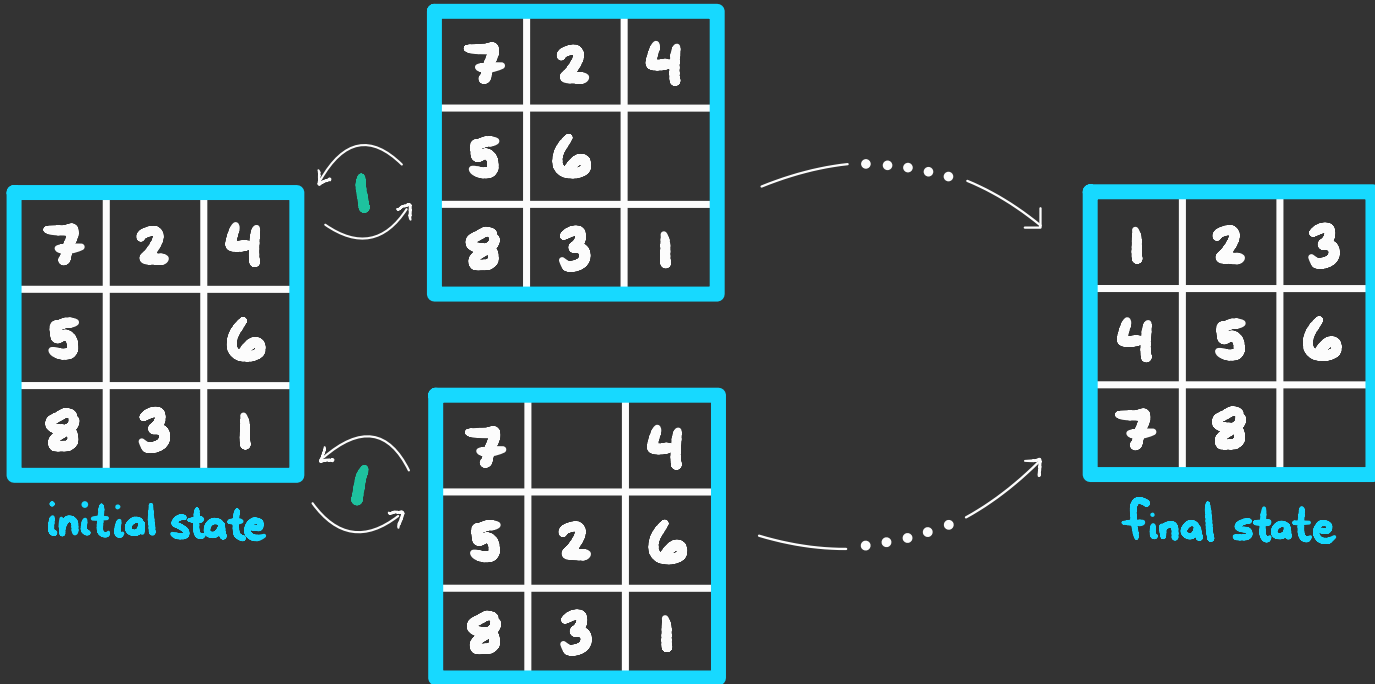


admissible
heuristics

CSCI
270

how do we design
admissible / consistent
heuristics?

consider sliding tile puzzles



what might be an
admissible heuristic?

$$H(q) \leq H^*(q)$$

7	2	4
5		6
8	3	1

initial state

heuristic
function

$$H(q)$$

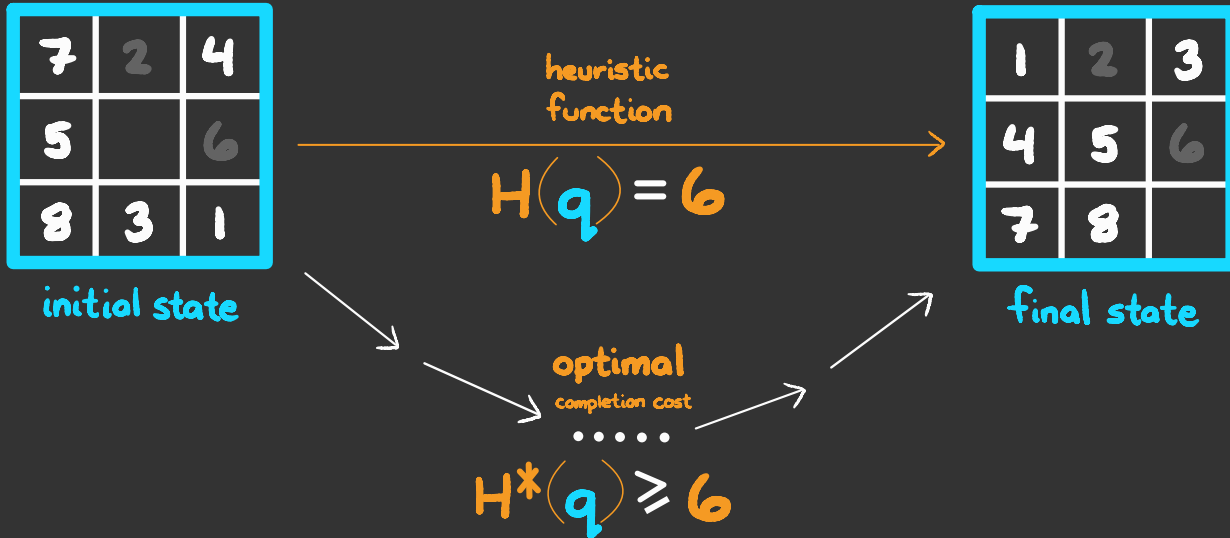
1	2	3
4	5	6
7	8	

final state

optimal
completion cost

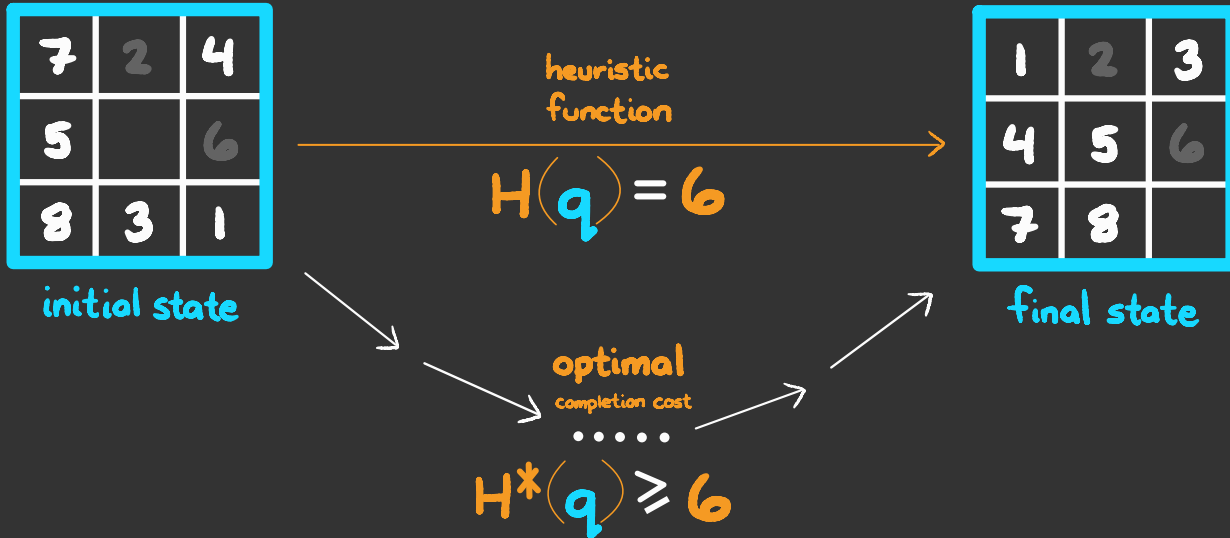
$$\dots\dots\dots H^*(q)$$

idea 1: number of misplaced tiles



why?
admissible

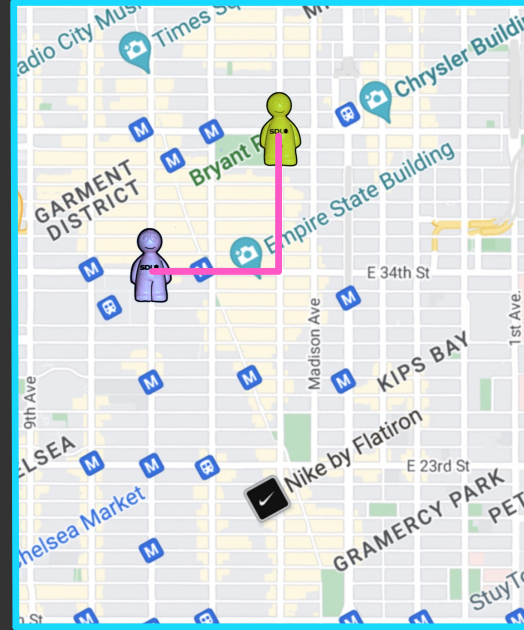
idea 1: number of misplaced tiles



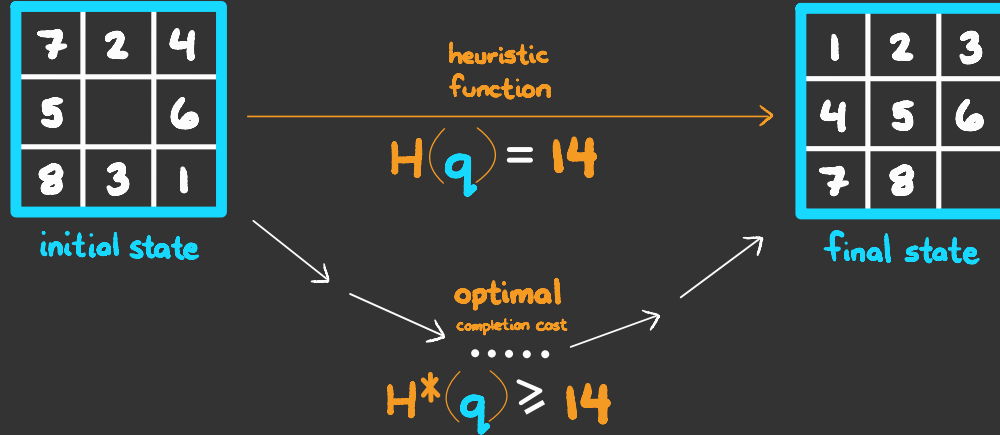
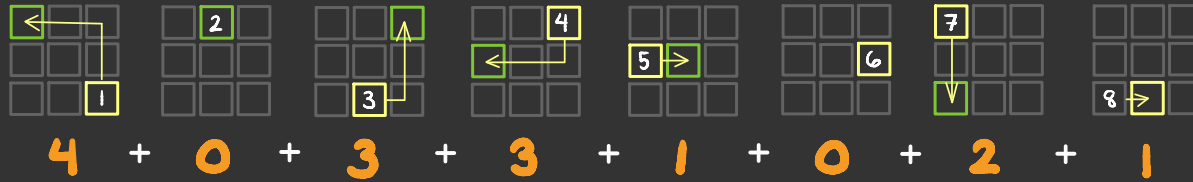
why?
admissible

each move displaces exactly one tile

idea 2: manhattan distance

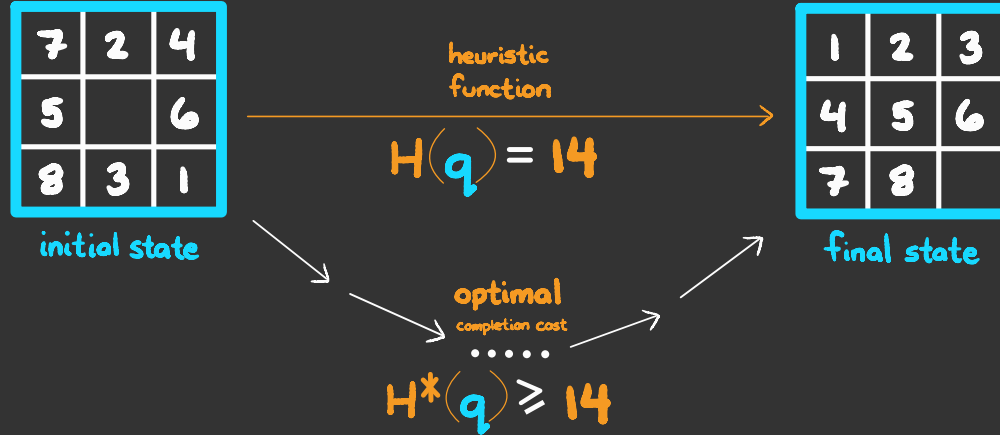
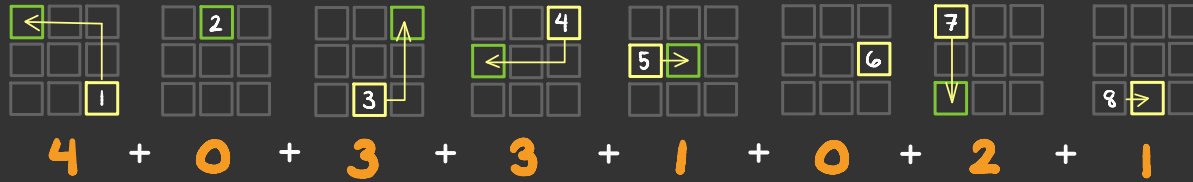


idea 2: cumulative manhattan distance between current tile locations and final tile locations



why?
admissible

idea 2: cumulative manhattan distance between current tile locations and final tile locations



why?
admissible

each move displaces exactly one tile horizontally or vertically

idea 2b : cumulative manhattan distance between current tile locations and final tile locations + empty space



7	2	4
5		6
8	3	1

initial state

heuristic
function

$$H(q) = 16$$

1	2	3
4	5	6
7	8	

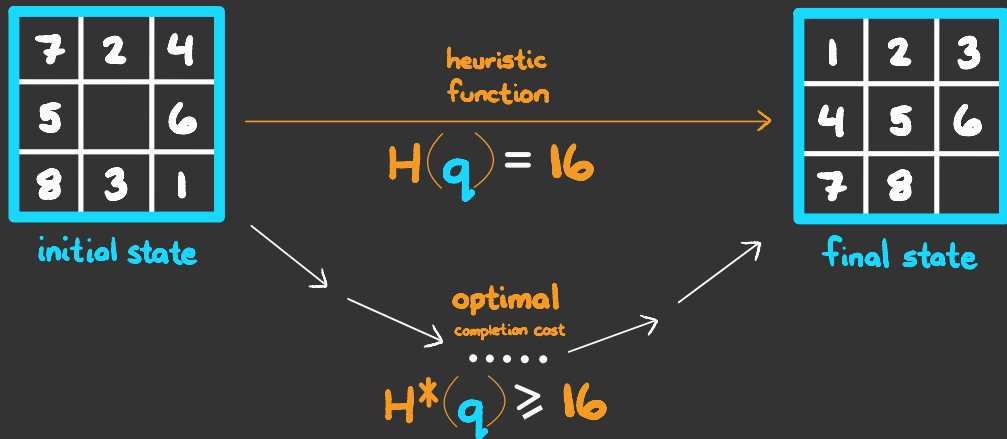
final state

optimal
completion cost

$$H^*(q) \geq 16$$

why?
admissible

idea 2b : cumulative manhattan distance between current tile locations and final tile locations + empty space

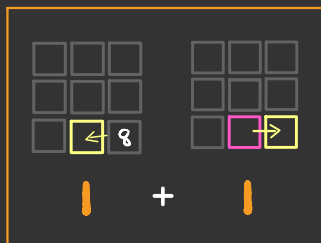


why?
admissible

it is not

1	2	3
4	5	6
7		8

initial state



heuristic
function

$$H(q) = 2$$

1	2	3
4	5	6
7	8	

final state

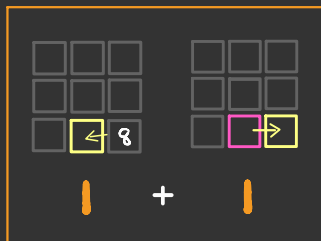
optimal
completion cost

...

$$H^*(q) \geq 2?$$

1	2	3
4	5	6
7		8

initial state



heuristic
function

$$H(q) = 2$$

1	2	3
4	5	6
7	8	

final state

optimal
completion cost

...

~~$$H^*(q) \geq 2?$$~~

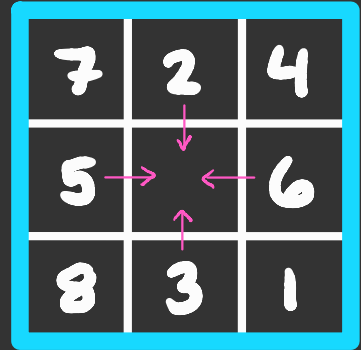
no. $H^*(q) = 1$

is there a less ethereal way
to derive admissible heuristics?



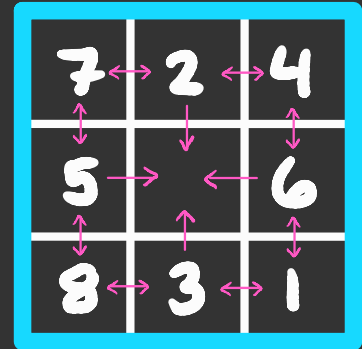
relax the problem definition!

in the eight-puzzle, a tile can
move from square A to square B
if A is adjacent to B and B is empty



relax the problem definition!

in the eight-puzzle, a tile can
move from square A to square B
if A is adjacent to B ~~and B is empty~~

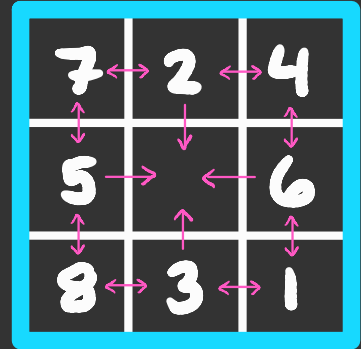


cost of the
optimal :
solution

?

relax the problem definition!

in the eight-puzzle, a tile can
move from square A to square B
if A is adjacent to B ~~and B is empty~~

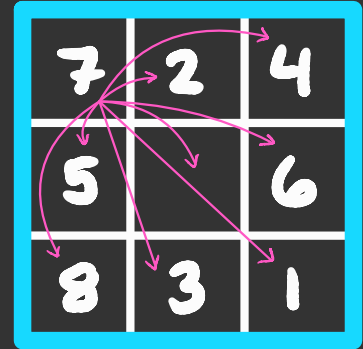


cost of the
optimal
solution

- cumulative manhattan distance between current tile locations and final tile locations

relax the problem definition!

in the eight-puzzle, a tile can
move from square A to square B
~~if A is adjacent to B and B is empty~~

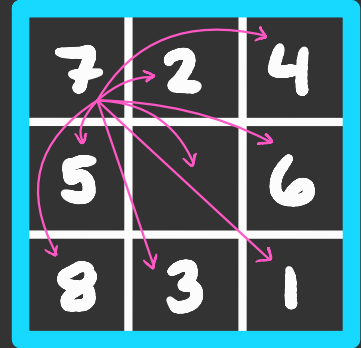


cost of the
optimal :
solution

?

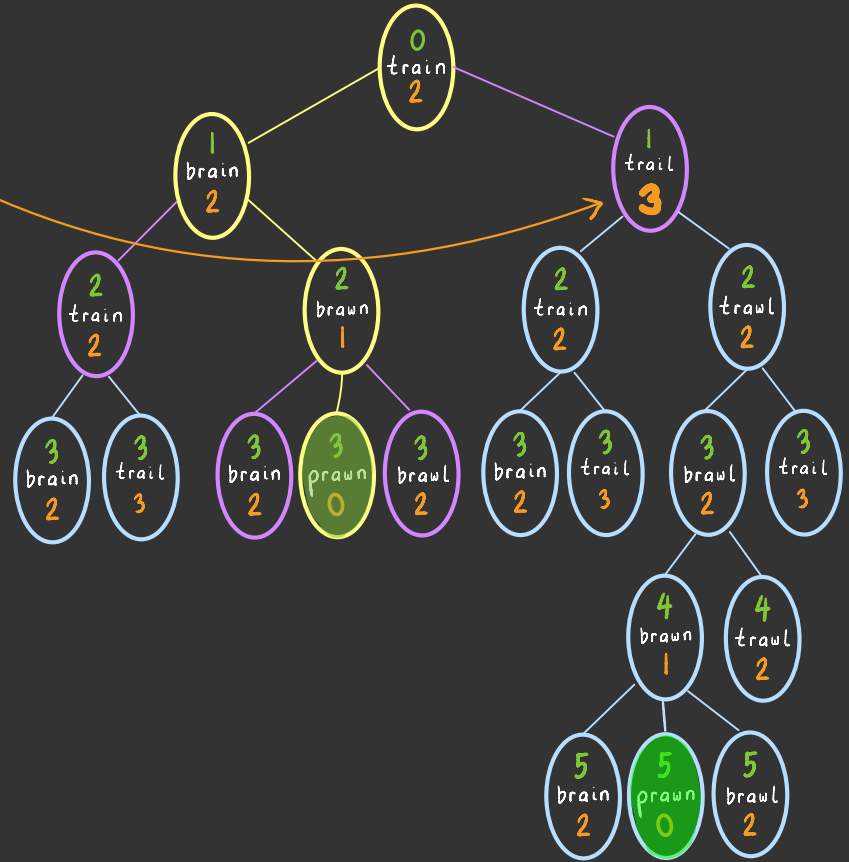
relax the problem definition!

in the eight-puzzle, a tile can
move from square A to square B
~~if A is adjacent to B and B is empty~~



cost of the
optimal : number of misplaced tiles
solution

what **heuristic**
did we use for
word ladder?

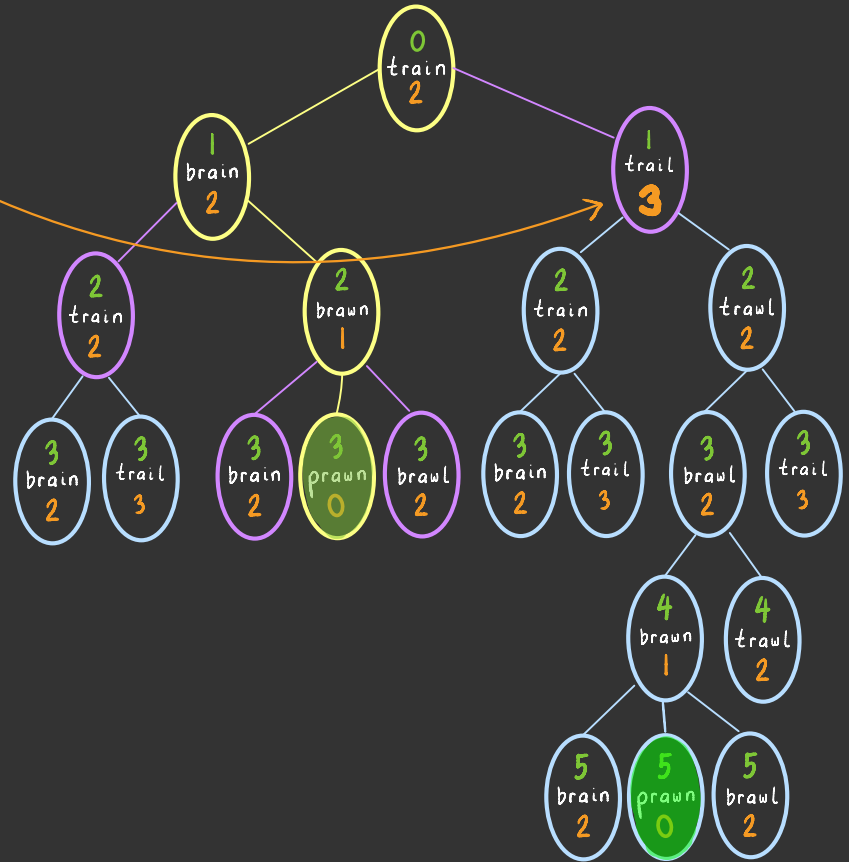


what **heuristic**
did we use for
word ladder?

t	r	a	i	l
---	---	---	---	---

number of letters
different from target word

p	r	a	w	n
---	---	---	---	---



relax the problem definition!

rules

of word ladder

- ▶ change **one** letter at a time
- ▶ each intermediate word must be valid english

your score is the number
of steps you take

p r a w n



t r a i n

cost of the
optimal :
solution

number of letters
different from target word

relax the problem definition!

rules

of word ladder

► change **one**
letter at a time

~~► each intermediate
word must be
valid english~~

your score is the number
of steps you take

p r a w n



t r a i n

cost of the
optimal :
solution

number of letters
different from target word